

Flights Merging Datasets In-Class Work

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NYCFLIGHTS13

```
require(nycflights13)
```

```
## Loading required package: nycflights13
```

```
## Warning: package 'nycflights13' was built under R version 4.4.3
```

```
data("flights")
data("weather")
head(flights)
```

```
## # A tibble: 6 x 19
```

```
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
## 1  2013     1     1     517           515           2     830           819
## 2  2013     1     1     533           529           4     850           830
## 3  2013     1     1     542           540           2     923           850
## 4  2013     1     1     544           545          -1    1004          1022
## 5  2013     1     1     554           600          -6     812           837
## 6  2013     1     1     554           558          -4     740           728
```

```
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dtm>
```

```
head(weather)
```

```
## # A tibble: 6 x 15
```

```
##   origin year month   day hour temp dewp humid wind_dir wind_speed wind_gust
##   <chr>  <int> <int> <int> <int> <dbl> <dbl> <dbl>   <dbl>    <dbl>    <dbl>
## 1 EWR    2013     1     1     1 39.0 26.1 59.4     270     10.4      NA
## 2 EWR    2013     1     1     2 39.0 27.0 61.6     250      8.06      NA
## 3 EWR    2013     1     1     3 39.0 28.0 64.4     240     11.5      NA
## 4 EWR    2013     1     1     4 39.9 28.0 62.2     250     12.7      NA
## 5 EWR    2013     1     1     5 39.0 28.0 64.4     260     12.7      NA
## 6 EWR    2013     1     1     6 37.9 28.0 67.2     240     11.5      NA
```

```
## # i 4 more variables: precip <dbl>, pressure <dbl>, visib <dbl>,
## #   time_hour <dtm>
```

```
flights_clean <- flights %>% select(year, month, day, dep_time, arr_time, carrier, flight, origin, dest)
weather_clean <- weather %>% select(year, month, day, temp, dewp, humid, wind_speed)
flights_weather <- flights_clean %>% left_join(weather_clean, by = c("year", "month", "day"))
```

```
## Warning in left_join(., weather_clean, by = c("year", "month", "day")): Detected an unexpected many-
```

```
## i Row 1 of `x` matches multiple rows in `y`.
## i Row 1 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
## "many-to-many"` to silence this warning.
```

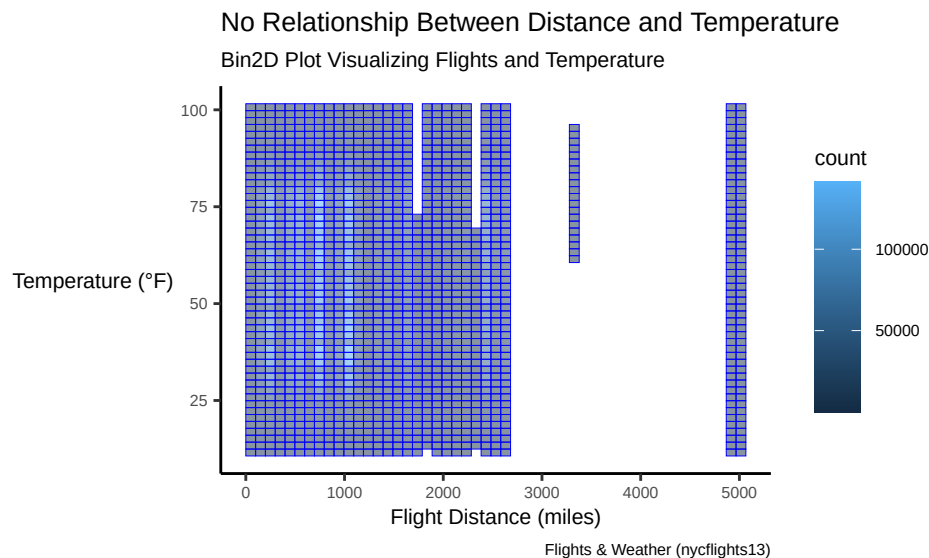
```
head(flights_weather)
```

```
## # A tibble: 6 x 16
##   year month   day dep_time arr_time carrier flight origin dest air_time
##   <int> <int> <int>   <int>   <int>   <chr>   <int> <chr> <chr>   <dbl>
## 1  2013     1     1     517     830    UA      1545 EWR   IAH     227
## 2  2013     1     1     517     830    UA      1545 EWR   IAH     227
## 3  2013     1     1     517     830    UA      1545 EWR   IAH     227
## 4  2013     1     1     517     830    UA      1545 EWR   IAH     227
## 5  2013     1     1     517     830    UA      1545 EWR   IAH     227
## 6  2013     1     1     517     830    UA      1545 EWR   IAH     227
## # i 6 more variables: distance <dbl>, dep_delay <dbl>, temp <dbl>, dewp <dbl>,
## #   humid <dbl>, wind_speed <dbl>
```

```
# Are longer flights more common in specific weather conditions?
```

```
gf_bin2d(temp ~ distance, data = flights_weather, bins = 50, color = "blue", alpha = 0.5) %>% gf_labs(x
```

```
## Warning: Removed 1766 rows containing non-finite outside the scale range
## (`stat_bin2d()`).
```



```
gf_bin2d(temp ~ dep_delay, data = flights_weather, bins = 50,color = "blue", alpha = 0.5) %>% gf_labs(x
```

```
## Warning: Removed 593500 rows containing non-finite outside the scale range
## (`stat_bin2d()`).
```

Temperature has an inverse relationship with Delay

Bin2D Plot Visualizing Departure Delays Across Temperatures

